POWER PULSE

Symmetrical (3-Phase) Faults Important Formulae

Notes by Balaji Sir

1 Fault current

1. $$I_f = \frac{V_{pf}}{Z_{th} + Z_f} = \frac{1}{Z_{th} + Z_f} \text{ pu}$$

pre-fault voltage 1.0 pu

2. $$I_f(\text{A}) = I_{base} \times I_{f,pu}, \qquad S_{sc} = \sqrt{3} V_L I_f = \frac{MVA_{base}}{X_{pu}}$$

3. $$I_{sc} = I_{rated} \times \frac{100}{\%X}$$

2 Machine behaviour

4. $$I'' = \frac{E''}{X''_d} > I' = \frac{E'}{X'_d} > I = \frac{E}{X_d}$$

subtransient > transient > steady state

5. $i(t) = \text{AC (decaying)} + \text{DC offset}; \quad \text{asymmetrical rms} \approx 1.6 \times \text{symmetrical}$

maximum offset

3 Z-bus method

6. $$I_f(k) = \frac{V_k(0)}{Z_{kk} + Z_f}, \qquad V_i(F) = V_i(0) - Z_{ik} I_f(k)$$

7. $$I_{ij} = \frac{V_i(F) - V_j(F)}{z_{ij}}$$

post-fault current in a line

4 Reactor and breaker

8. $X_{reactor} = X_{required} - X_{existing}$ (pu, on a common base)

current-limiting reactor

9. $\text{Breaking MVA} = \sqrt{3}\, V_{rated}\, I_b \qquad \text{Making current} \approx 2.55\, I_{sc,rms}$

10. Neglect loads, resistance and pre-fault currents (simple method); use sub-transient reactance